

Recomart End-to-End Data pipeline Orchstration job

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In [5]: from prefect import flow, task, get_run_logger
from prefect.artifacts import create_markdown_artifact # New import for UI reports
import subprocess
import sys

# Each task below wraps an external Python script to maintain modularity

@task(retries=2, retry_delay_seconds=30)
def run_ingestion():
    """Ingest data from CSV and JSON Streaming Files """
    logger = get_run_logger()
    logger.info("Starting Data Ingestion Stage...")
    result = subprocess.run([sys.executable, "scripts/ingestion.py"], capture_output=True, text=True)
    if result.returncode != 0:
        logger.error(f"Ingestion Failed: {result.stderr}")
        raise Exception("Ingestion script failed.")
    logger.info("Ingestion Success.")

@task(log_prints=True)
def run_validation():
    """Task 4: Data Profiling and Validation with GX Output Logging"""
    logger = get_run_logger()
    logger.info("Starting Data Validation Stage...")

    # Run the script and capture output
    result = subprocess.run(
        [sys.executable, "scripts/dq_validation.py"],
        capture_output=True,
        text=True
    )

    # 1. Print the stdout to the Prefect Logs
    if result.stdout:
        print("--- Great Expectations Standard Output ---")
        print(result.stdout)

    # 2. Handle failures and print errors
    if result.returncode != 0:
        logger.error("Great Expectations Validation Failed!")
        if result.stderr:
            print("--- GX Error Log ---")
            print(result.stderr)

    # 3. Create a Markdown Artifact to highlight the failure in the UI
    create_markdown_artifact(
        key="gx-validation-failure",
        markdown=f"""## ❌ Validation Failed\nCheck the task logs for full trace.\n\n**Error Summary:**""",
        description="High-level failure notice for Data Quality"
    )
    raise Exception("Data validation failed.")

    logger.info("Validation Success.")

    # 4. Create a Success Artifact
    create_markdown_artifact(
        key="gx-validation-success",
        markdown=f"""## ✅ Validation Passed\nGreat Expectations found no critical data quality issues.""",
        description="Data Quality Check Status"
    )

@task(log_prints=True)
def run_preparation():
    """Task 5: Data Preparation and Cleaning """
    logger = get_run_logger()
    logger.info("Starting Data Preparation Stage...")
    subprocess.run([sys.executable, "scripts/preparation.py"], check=True)
    logger.info("Preparation Success.")

@task(log_prints=True)
def run_feature_engg():
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"""Task 6 & 7: Feature Engineering and Feature Store update """
logger = get_run_logger()
logger.info("Updating Feature Engineering...")
subprocess.run([sys.executable, "scripts/feature_engineering.py"], check=True)
logger.info("Feature Engineering updated successfully.")

@task(log_prints=True)
def run_feature_store():
    """Task 6 & 7: Feature Store update """
    logger = get_run_logger()
    logger.info("Updating Feature Store...")
    subprocess.run([sys.executable, "scripts/feature_store.py"], check=True)
    logger.info("Feature Store updated successfully.")

@task(log_prints=True)
def run_schema_validator():
    """Task 6 & 7: Schema validation and versioning """
    logger = get_run_logger()
    logger.info("Validating schema & versioning...")
    subprocess.run([sys.executable, "scripts/feature_validator.py"], check=True)
    logger.info("Feature Store updated successfully.")

@task(log_prints=True)
def run_model_training():
    """Task 9: Model Training and MLflow tracking """
    logger = get_run_logger()
    logger.info("Starting Model Training Stage...")
    subprocess.run([sys.executable, "scripts/ml_model_training.py"], check=True)
    logger.info("Model Training and Experiment Tracking Complete.")

@flow(name="RecoMart_End_to_End_Orchestration", log_prints=True)
def recomart_full_pipeline():
    """Main flow to orchestrate Task """
    # Define execution order
    ingest = run_ingestion()
    validate = run_validation(wait_for=[ingest])
    prep = run_preparation(wait_for=[validate])
    feateng = run_feature_engg(wait_for=[prep])
    feat = run_feature_store(wait_for=[feateng])
    ver = run_schema_validator(wait_for=[feat])
    run_model_training(wait_for=[ver])

if __name__ == "__main__":
    recomart_full_pipeline()

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12:55:28.087 | INFO | Flow run 'jumping-raven' - Beginning flow run 'jumping-raven' for flow 'RecoM
12:55:28.155 | INFO | Task run 'run_ingestion-45f' - Starting Data Ingestion Stage...
12:56:00.580 | INFO | Task run 'run_ingestion-45f' - Ingestion Success.
12:56:00.594 | INFO | Task run 'run_ingestion-45f' - Finished in state Completed()
12:56:00.616 | INFO | Task run 'run_validation-fb2' - Starting Data Validation Stage...
12:56:14.578 | INFO | Task run 'run_validation-fb2' - --- Great Expectations Standard Output ---

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12:56:14.582 | INFO | Task run 'run_validation-fb2' - Validation Success: False
Stats: {'evaluated_expectations': 9, 'successful_expectations': 8, 'unsuccessful_expectations': 1, 'su

Column 'action' failed!
Issue: Unknown Expectation
Unexpected count: 1233
Sample invalid data: ['No Action', 'No Action', 'No Action', 'No Action', 'No Action', 'No Action', 'N
Success: 10233 rows
Failed: 1233 rows

Sample of Invalid Records:

	user_id	product_id	action	...	brand	price	interaction_score
8244	USER_345	rm_100	No Action	...	Reliance	743.75	NaN
8245	USER_345	rm_100	No Action	...	Samsung	472.73	NaN
8246	USER_345	rm_100	No Action	...	Unknown	1153.68	NaN
8247	USER_345	rm_100	No Action	...	Unknown	1105.95	NaN
8248	USER_345	rm_100	No Action	...	Brand ABC	307.89	NaN

[5 rows x 8 columns]
Rows before: 11466 , Rows after cleansing: 10233

Successfully loaded 10233 rows to postgres table 'userinteraction'

12:56:14.589 | INFO | Task run 'run_validation-fb2' - Validation Success.
12:56:14.754 | INFO | Task run 'run_validation-fb2' - Finished in state Completed()
12:56:14.772 | INFO | Task run 'run_preparation-44a' - Starting Data Preparation Stage...
12:56:26.856 | INFO | Task run 'run_preparation-44a' - Preparation Success.
12:56:26.864 | INFO | Task run 'run_preparation-44a' - Finished in state Completed()
12:56:26.878 | INFO | Task run 'run_feature_engg-7b5' - Updating Feature Engineering...
12:56:40.444 | INFO | Task run 'run_feature_engg-7b5' - Feature Engineering updated successfully.
12:56:40.451 | INFO | Task run 'run_feature_engg-7b5' - Finished in state Completed()
12:56:40.467 | INFO | Task run 'run_feature_store-0de' - Updating Feature Store...
12:56:42.903 | INFO | Task run 'run_feature_store-0de' - Feature Store updated successfully.
12:56:42.913 | INFO | Task run 'run_feature_store-0de' - Finished in state Completed()
12:56:42.933 | INFO | Task run 'run_schema_validator-a9b' - Validating schema & versioning...
12:56:45.890 | INFO | Task run 'run_schema_validator-a9b' - Feature Store updated successfully.
12:56:45.900 | INFO | Task run 'run_schema_validator-a9b' - Finished in state Completed()
12:56:45.916 | INFO | Task run 'run_model_training-8da' - Starting Model Training Stage...
12:57:00.310 | INFO | Task run 'run_model_training-8da' - Model Training and Experiment Tracking Co

12:57:00.316 | INFO | Task run 'run_model_training-8da' - Finished in state Completed()
12:57:00.479 | INFO | Flow run 'jumping-raven' - Finished in state Completed()

In []: